

System Design - Sequence Diagrams

This section presents the sequence diagrams for the Fitness Tracker application. Each diagram corresponds to one use case from the use-case diagram and traces the message flow between the actor and the participating objects: MainGUI, AddGUI, EditGUI, SearchGUI, ActivityService, and StorageManager.

Notation: object boxes label each participant at the top of the diagram, vertical dashed lines are the lifelines, narrow vertical rectangles are activation bars, solid arrows are synchronous calls (with the «create» stereotype on object instantiation), and dashed arrows are returns.

Sequence Diagram 1 - Add New Activity

UC1: Add Activity

The user opens the Functions dropdown on the main toolbar and chooses Add. MainGUI creates an AddGUI dialog, the user fills in the activity fields and presses Save. AddGUI validates the input, calls `addActivity()` on `ActivityService`, which in turn asks `StorageManager` to persist the updated list. Once the save returns successfully, AddGUI invokes the `onSave` callback so MainGUI refreshes its display and the new entry becomes visible.

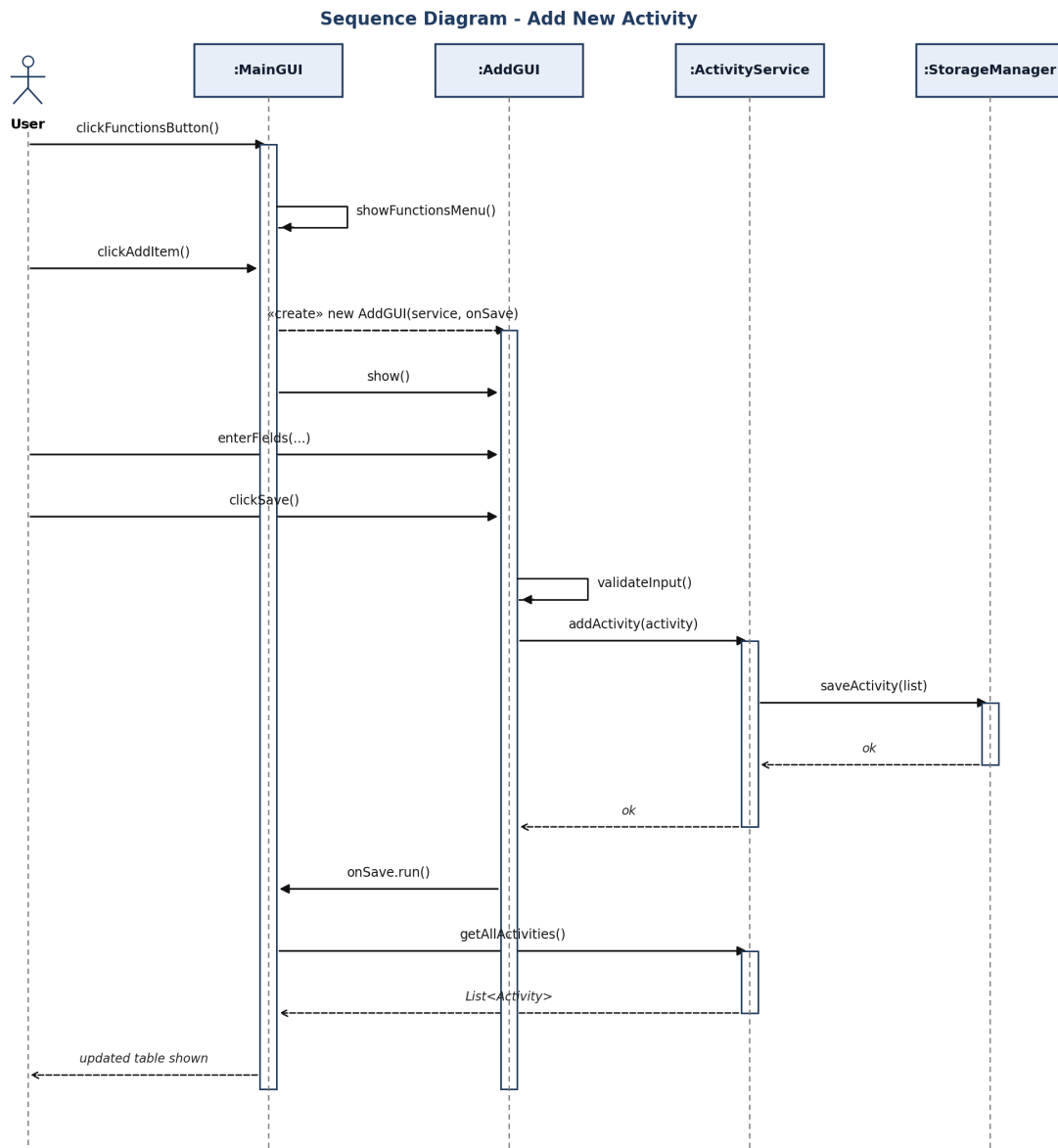


Figure 1 - Add New Activity

Sequence Diagram 2 - View Activities

UC2: View Activities

On launch or refresh, MainGUI asks ActivityService for the current list, which is loaded from StorageManager. MainGUI then populates its scrollable table so the user can browse all saved activities.

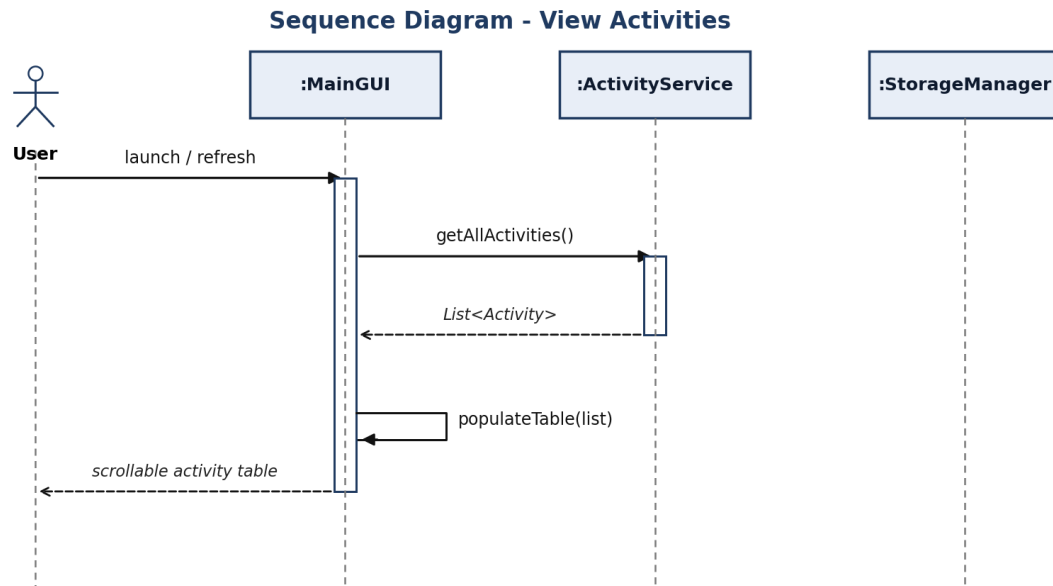


Figure 2 - View Activities

Sequence Diagram 3 - Search Activities

UC3: Search Activities

The user opens the Functions dropdown on the main toolbar and chooses Search. MainGUI creates a SearchGUI dialog and the user selects a category (Type, Date, Collaborator, or Notes). SearchGUI swaps the input widget to fit the chosen category. Once the user enters a value and presses Search, SearchGUI delegates the filtering to ActivityService.search(), which returns the matching subset for display.

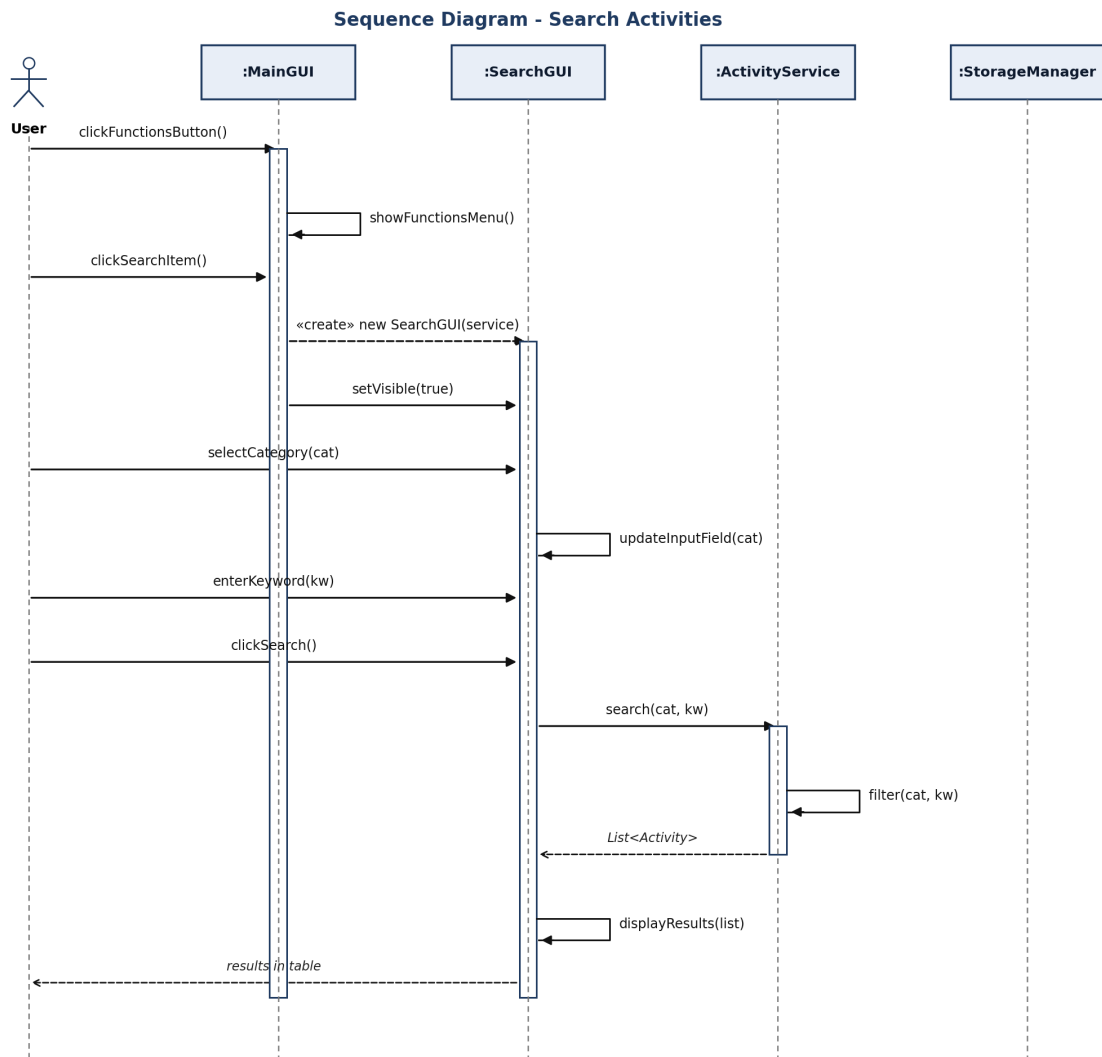


Figure 3 - Search Activities

Sequence Diagram 4 - Edit Activity

UC4: Edit Activity (extends View / Search)

The user right-clicks a row to open the popup menu and chooses Edit. MainGUI passes the selected activity into a new EditGUI, which displays the existing values for the user to modify. After validation, EditGUI calls updateActivity() on ActivityService, persistence completes, and the onSave callback runs to refresh MainGUI.

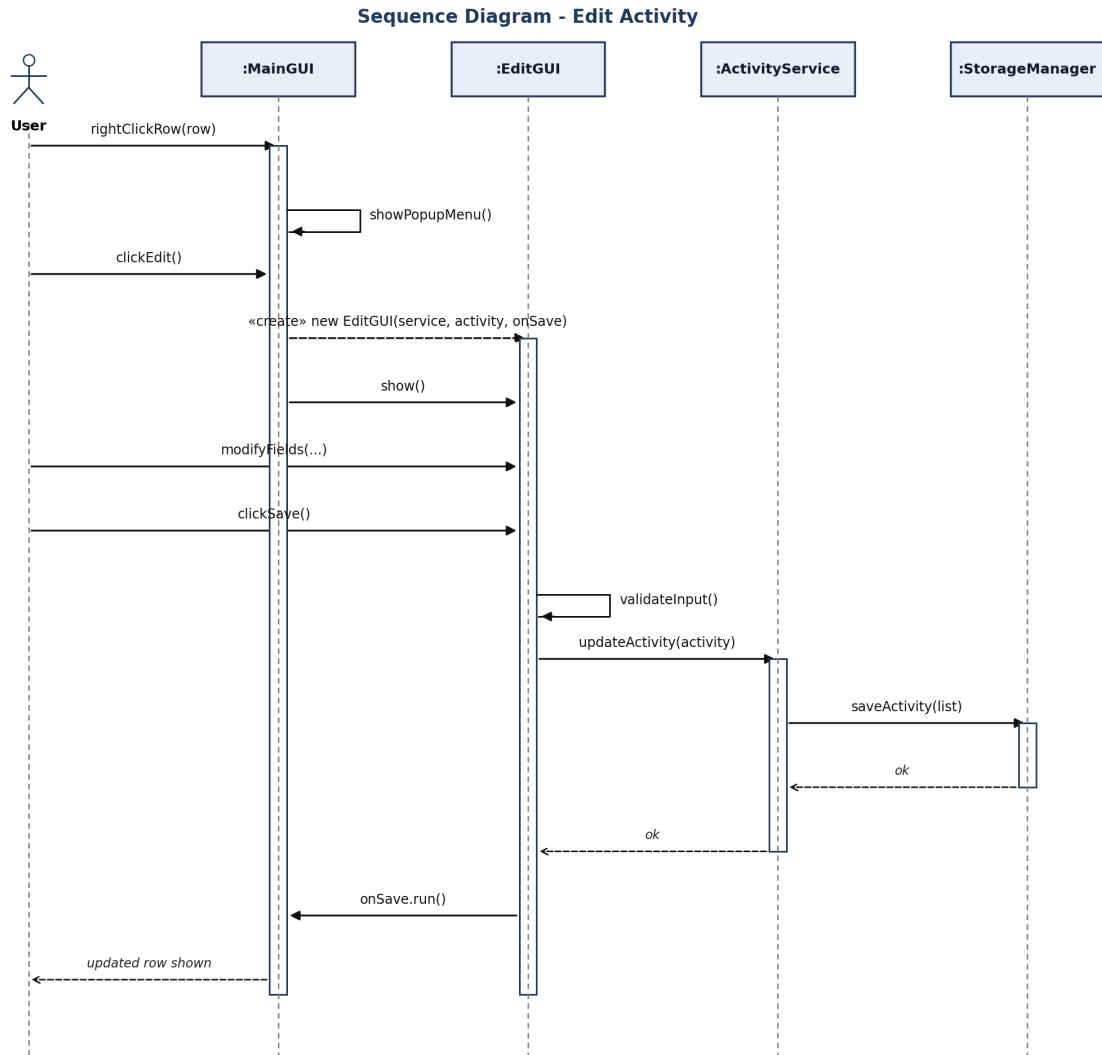


Figure 4 - Edit Activity

Sequence Diagram 5 - Delete Activity

UC5: Delete Activity (extends View / Search)

The user right-clicks a row, picks Delete, and confirms in the dialog. MainGUI calls deleteActivity() on ActivityService with the activity's id, the service removes the record and asks StorageManager to persist the updated list. MainGUI refreshes the table to reflect the removal.

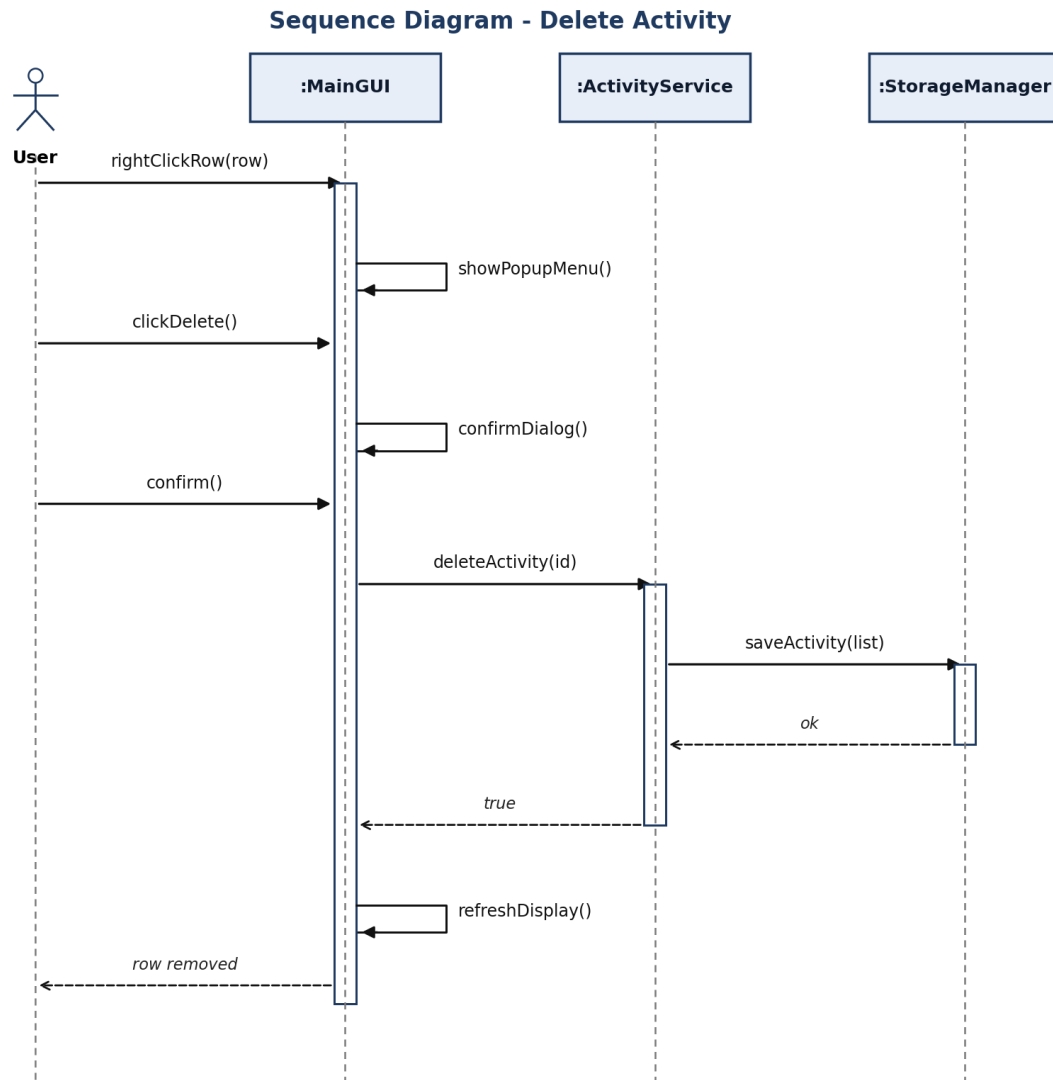


Figure 5 - Delete Activity